Submission PY1-MID-SUB01

Question PY1-MID-Q01
scratch: show step
1. range(4) gives 0 through 3
2. keep even i: 0, 2
3. squares => [0, 4]
=> FINAL: [0, 4] (check units/interpretation)

Question PY1-MID-Q02
scratch: show step
1. counts = Counter(xs)
2. for x in xs: if counts[x] == 1: return x
3. return None; two linear passes $\Rightarrow$ O(n) time, O(n) space
=> FINAL: scan counts, then scan in order; return first count==1 else None; O(n) time, O(n) space (check units/interpretation)

Question PY1-MID-Q03
scratch: show step
1. A single pass fixes only adjacent inversions encountered once.
2. Repeat until no swaps (or n–1 passes), shrinking the suffix after each pass.
=> FINAL: repeat passes or track the shrinking unsorted suffix; one pass alone is not a full sort (check units/interpretation)

Question PY1-MID-Q04
scratch: show step
1. linear scan: up to 64 comparisons, $\Theta(n)$
2. binary search: at most ■log■(64)■+1 comparisons, $\Theta(\log n)$
=> FINAL: linear $\Theta(n)$, up to 64; binary $\Theta(\log n)$, about 7 (check units/interpretation)

Question PY1-MID-Q05
scratch: show step
1. Removing shifts later elements left while the iterator advances.
2. Build a filtered list or iterate over a copy, then replace the original.
=> FINAL: mutation shifts elements and skips checks; use a filtered copy or iterate over a copy (check units/interpretation)

Question PY1-MID-Q06
scratch: show step
1. range(4) gives 0 through 3
2. keep even i: 0, 2
3. squares => [0, 4]
=> FINAL: [0, 4] (check units/interpretation)

Question PY1-MID-Q07

scratch: show step

1. counts = Counter(xs)
2. for x in xs: if counts[x] == 1: return x
3. return None; two linear passes ⇒ O(n) time, O(n) space

=> FINAL: scan counts, then scan in order; return first count==1 else None; O(n) time, O(n) space (check units/interpretation)

Question PY1-MID-Q08

scratch: show step

1. A single pass fixes only adjacent inversions encountered once.
2. Repeat until no swaps (or n–1 passes), shrinking the suffix after each pass.

=> FINAL: repeat passes or track the shrinking unsorted suffix; one pass alone is not a full sort (check units/interpretation)